

Lecture 1: Introduction and Data Example

Deep Learning for Actuarial Modeling
Summer School 'Lifelong Learning'
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ABSTRACT

We start by introducing a car insurance pricing problem which is the most typical example for applying regression modeling and machine learning tools. This example will serve us as a running example through many of the chapters and lectures.

1 Actuarial modeling – a pricing problem

- We introduce a car insurance pricing problem.
- This pricing problem is based on *claims frequencies*.
- The analysis in this first lecture will mostly be empirical.
- It considers the popular French motor third party liability (MTPL) claims dataset `freMTPL2freq` of Dutang, Charpentier and Gallic (2024).
- Generally, raw data needs *data cleaning*, e.g., correcting for data errors, inputting missing values, merging partial information, etc.
- The data presented here has already been cleaned; for the data cleaning applied see Appendix B in Wüthrich and Merz (2023).

1.1 Load and illustrate the French MTPL data

```
load(file="../Data/freMTPL2freqClean.rda")      # there are other data
                                                formats
dat <- freMTPL2freqClean
head(dat[,c(1:11,13)])
```

- The (cleaned) data contains the following variables:
 - `IDpol` is a unique policy identifier
 - `Exposure` is the time exposure (in yearly units)
 - `Area` is an area code (ordinal categorical)
 - `VehPower` is the power of the vehicle
 - `VehAge` is the age of the vehicle
 - `DrivAge` is the age of the policyholder
 - `BonusMalus` is the bonus-malus level of the policyholder
 - `VehBrand` indicates the vehicle brand (categorical)
 - `VehGas` indicates the gasoline type (categorical)
 - `Density` is the population density at the living place of the policyholder
 - `Region` gives the French department of the license plate (categorical)
 - `ClaimNb` gives the observed number of claims
-

	IDpol	Exposure	Area	VehPower	VehAge	DrivAge	BonusMalus	VehBrand
VehGas								
1	4156370	0.06	D	6	6	20	100	B2
Regular								
2	4006798	0.29	E	6	7	29	59	B12
Diesel								
3	6084964	0.46	C	7	10	27	68	B1
Diesel								
4	2228865	0.08	D	4	15	34	50	B2
Regular								
5	4141911	1.00	A	5	22	44	50	B3
Diesel								
6	1103292	0.60	C	5	2	25	90	B5
Regular								
Density	Region	ClaimNb						
1	525	R82	0					
2	2498	R72	0					
3	123	R82	0					
4	1109	R24	0					
5	34	R72	0					
6	129	R41	0					

- For simplicity, we focus on the claims counts `ClaimNb` ; there would also be data available with claim amounts.

1.2 Claims count modeling and frequencies

- The above portfolio has $n = 678,007$ insurance policyholders. We label them by $i \in \{1, \dots, n\}$.
- These policyholders are characterized by *covariates* $\mathbf{X}_i$ describing their *policy features* such as the age of the car or the vehicle brand.
- For each policyholder i , there is an observed *number of claims* $N_i \geq 0$ within the given calendar year.
- The *time exposures* $v_i \in (0, 1]$ give the proportions of the calendar year during which the policies have been active (exposed).
- To account for the different time exposures, one typically studies and models the *claims frequencies*

$$Y_i = N_i/v_i.$$

- The general task in actuarial pricing is to *predict (forecast)* the (future) claims frequencies Y_i as accurately as possible.
- These predictions should take into account all available/relevant policyholder information $\mathbf{X}_i$, as this information may reveal structural differences in their propensity to claims.
- (Best) predictors are typically computed by conditional means

$$\mu(\mathbf{X}_i) := \mathbb{E}[Y_i | \mathbf{X}_i].$$

This is the expected claim of a policyholder with covariates $\mathbf{X}_i$.

- *Actuarial pricing* one aims at finding an accurate *regression function*

$$\mathbf{X} \mapsto \mu(\mathbf{X}),$$

that reflects the expected claim as a function of the covariates $\mathbf{X}$.

1.3 Regression modeling

- In general, the true data generating mechanism/model is *unknown*.
- Therefore, one cannot compute the regression function $\mathbf{X} \mapsto \mu(\mathbf{X})$.
- Thus, one needs to *estimate* μ from *past data*

$$\mathcal{L} = (Y_i, \mathbf{X}_i, v_i)_{i=1}^n.$$

- $\mathcal{L}$ is called *learning sample* because it is used to learn the unknown regression function $\mathbf{X} \mapsto \mu(\mathbf{X})$.
- Before solving this estimation/learning problem, one should always empirically study the available data $\mathcal{L} = (Y_i, \mathbf{X}_i, v_i)_{i=1}^n$.
- This empirical study is crucial to fully understand the problem and to select a suitable class of *candidate regression models* $\mathcal{M} = \{\mu\}$.

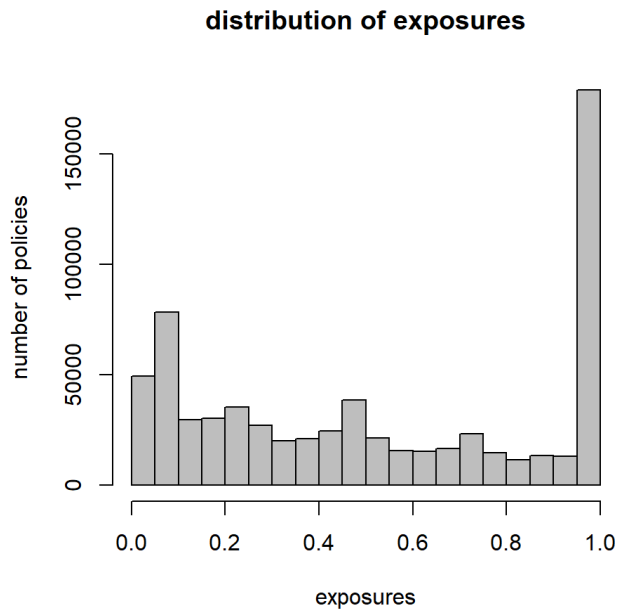
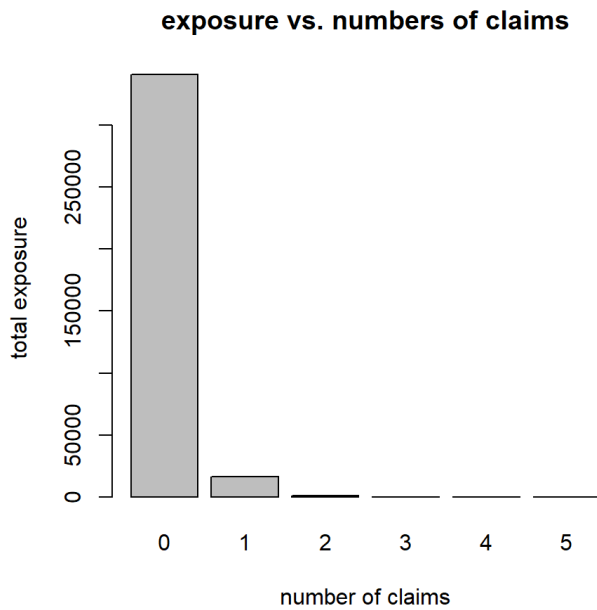
2 Empirical data analysis

- Before we try to find a suitable regression function $\mathbf{X} \mapsto \mu(\mathbf{X})$, we empirically analyze the learning sample $\mathcal{L}$.
- Such an empirical analysis of the learning sample $\mathcal{L}$ includes different *sample statistics* as well as *visualizations* of the data.
- For simplicity, we only consider the two covariates `DrivAge` and `Density` in this preliminary analysis. Later, all covariates are considered.
- We start by classifying the policies w.r.t. their numbers of claims.

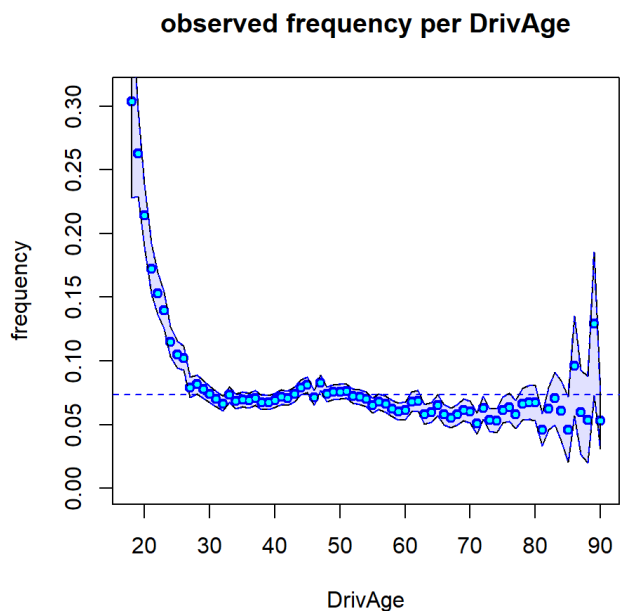
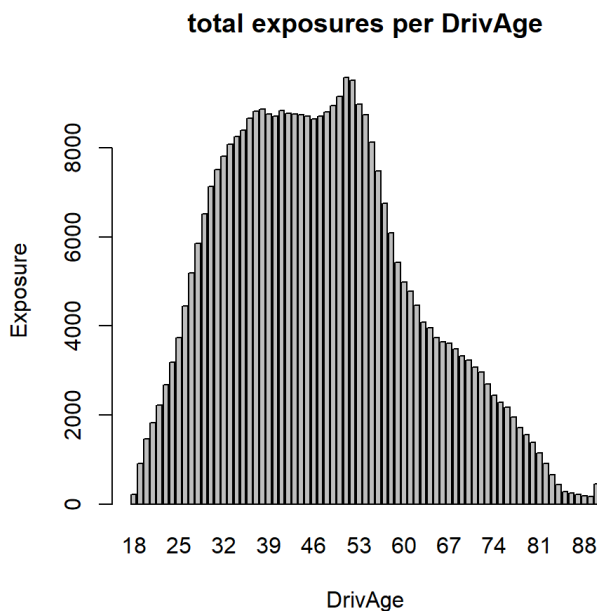
```
#  
library(arrow)  
library(tidyverse)  
#  
dat %>% group_by(ClaimNb) %>% summarize(Exposure = sum(Exposure))
```

	ClaimNb	Exposure
1	0	341090
2	1	16315
3	2	909
4	3	42
5	4	2
6	5	1

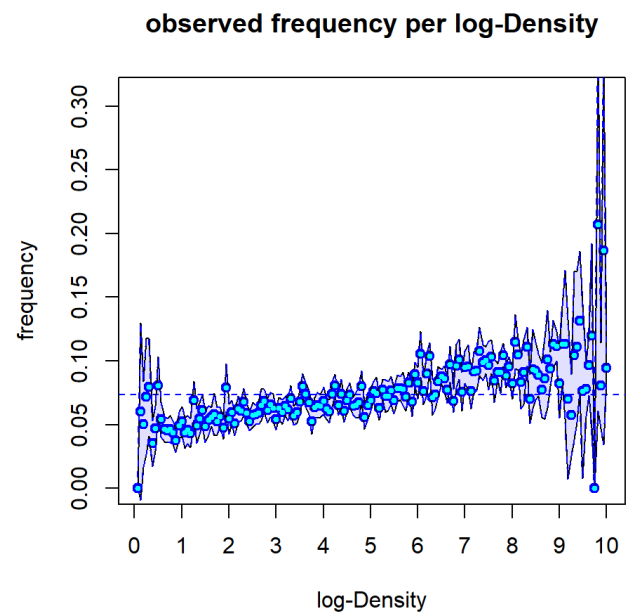
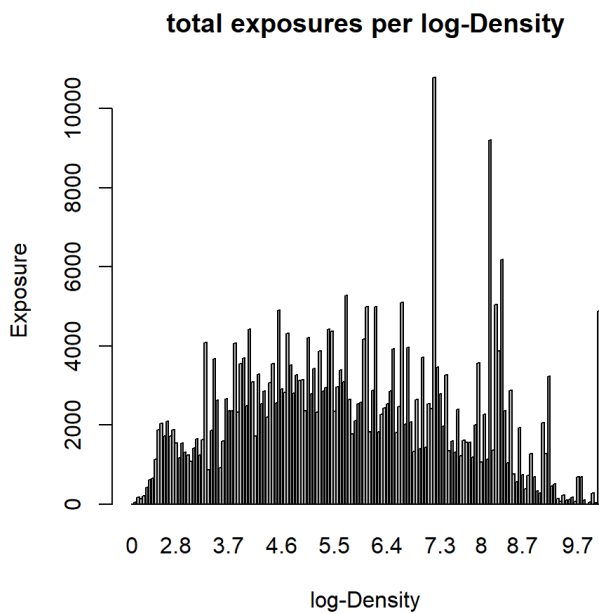
- The big majority of insurance policies does not suffer any claims.
- The empirical claims frequency is $\sum_{i=1}^n N_i / \sum_{i=1}^n v_i = 7.36\%$.
- In machine learning, this is called a *class imbalance problem*.
- In actuarial modeling, this is just the typical problem actuaries have to deal with, i.e., insurance claims are rare events.
- Thus, actuaries try to find *systematic structure* in data in which claims are rare events.



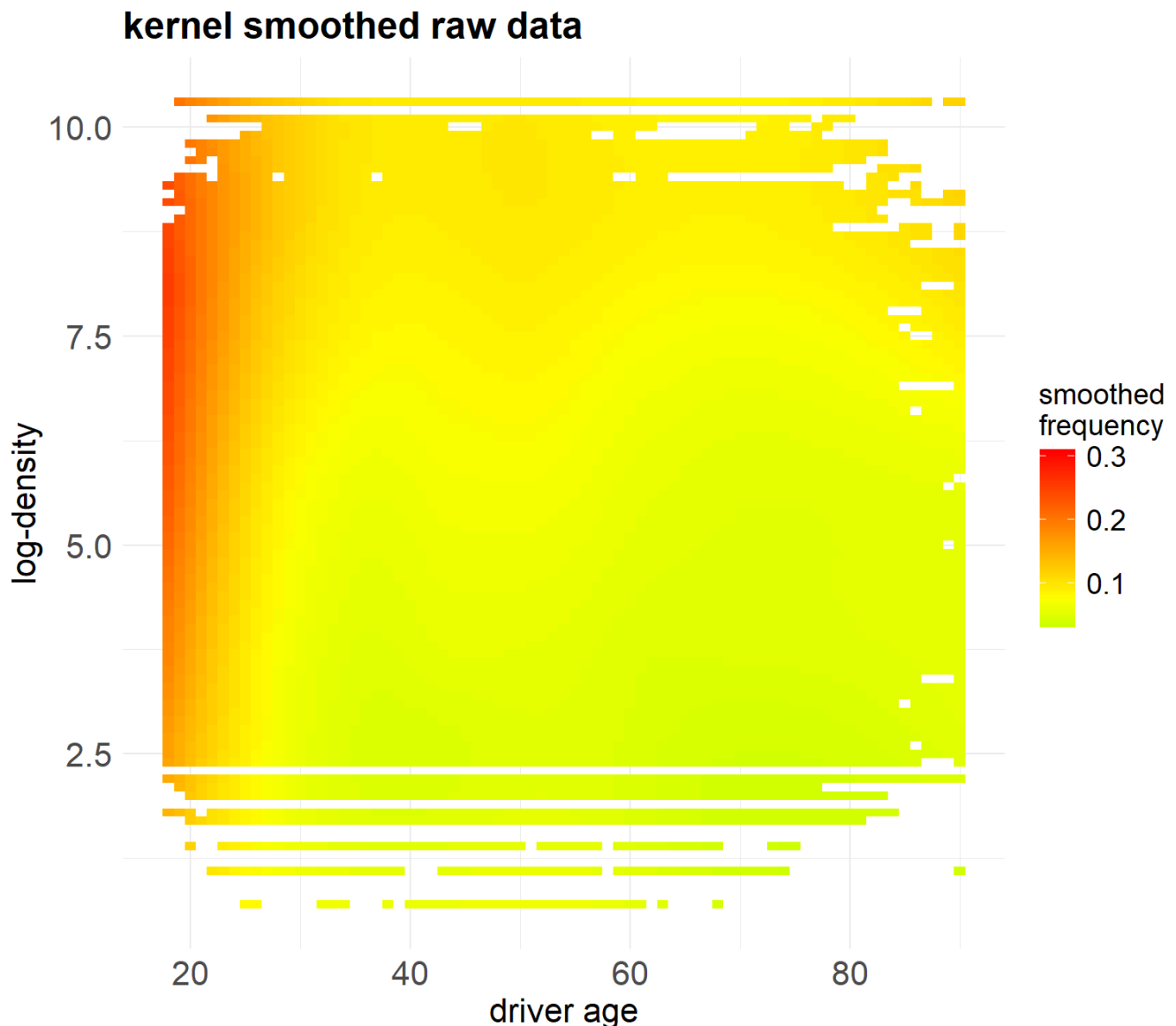
- (lhs): Class imbalance problem mentioned above.
- (rhs): Distribution of exposures $(v_i)_{i=1}^n$, roughly 25% of the policies have a full-year exposure $v_i = 1$ in this data.



- (lhs): Total exposure $\sum_i v_i \mathbf{1}_{\{\text{age}_i = \text{age}\}}$ per driver age (**DrivAge**).
- (rhs): Empirical frequency $\sum_i N_i \mathbf{1}_{\{\text{age}_i = \text{age}\}} / \sum_i v_i \mathbf{1}_{\{\text{age}_i = \text{age}\}}$ per **DrivAge** ; confidence bounds correspond to 2 std in a Poisson model.



- (lhs): Total exposure $\sum_i v_i \mathbf{1}_{\{LD_i=LD\}}$ per log-Density (class LD).
 - (rhs): Empirical frequency $\sum_i N_i \mathbf{1}_{\{LD_i=LD\}} / \sum_i v_i \mathbf{1}_{\{LD_i=LD\}}$ per log-Density; confidence bounds correspond to 2 std in a Poisson model.
-
- The previous plots show the *marginal distributions* of the claims.
 - Generally, covariates *interact*, and one cannot simply multiply marginal frequencies to receive the correct prices.
 - For example:
 - Young drivers may more likely live in the highly populated cities.
 - Thus, high frequencies at younger ages and in densely populated areas may be driven by the same risk factor.
 - Multiplying marginals will, thus, result in a double penalization of the same risk factor.
 - For this reason, we need to understand the *multivariate structure* to receive a good pricing functional $\mathbf{X} \mapsto \mu(\mathbf{X})$.
 - In other words, we need to estimate $\mu(\mathbf{X})$ *jointly* in the components of $\mathbf{X} = (X_1, \dots, X_q)^\top$, and *not marginally*.



- The above heatmap shows the empirical frequency as a function of the two covariates `DrvAge` and `log-Density`.
- The empirical data has been smoothed with the kernel smoothing method called `loess` (locally estimated scatter plot smoothing).
- The colored dots show the `loess` results in the covariate combinations that exist in our portfolio, and the white space corresponds to covariate combinations that are not available in our portfolio.
- The coloring shows quite some non-linear structure in the covariate components, and it does not support the pattern of multiplying marginals.

- This requires regression modeling or machine learning solutions, e.g.,
 - generalized linear models (GLMs)
 - neural networks and deep learning
 - regression trees
 - gradient boosting machines (GBMs)

3 A generalized linear model example

- We start with *generalized linear models* (GLMs) for the class $\mathcal{M}$ of the considered regression functionals $\mathbf{X} \mapsto \mu(\mathbf{X})$.
- The underlying mathematical theory is going to be presented in careful detail in the next lecture.
- The most simple GLM setting considers

$$\mathbf{X} \mapsto \mu(\mathbf{X}) = \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2),$$

for $\mathbf{X} = (X_1, X_2) = (\text{DrivAge}, \log\text{-Density})$ and parameter $(\beta_0, \beta_1, \beta_2)$.

```
### Model GLM 1 (Poisson for counts with offset and canonical log-link)
d.glm1 <- glm(ClaimNb ~ DrivAge + lDensity,
              data=dat, offset=log(Exposure), family=poisson())
#
dat$GLM1 <- fitted(d.glm1)
summary(d.glm1)
```

Call:

```
glm(formula = ClaimNb ~ DrivAge + lDensity, family = poisson(),  
     data = dat1, offset = log(Exposure))
```

Coefficients:

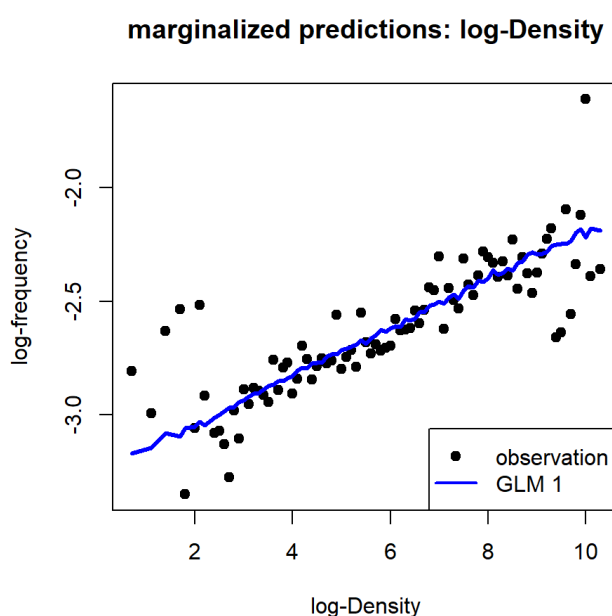
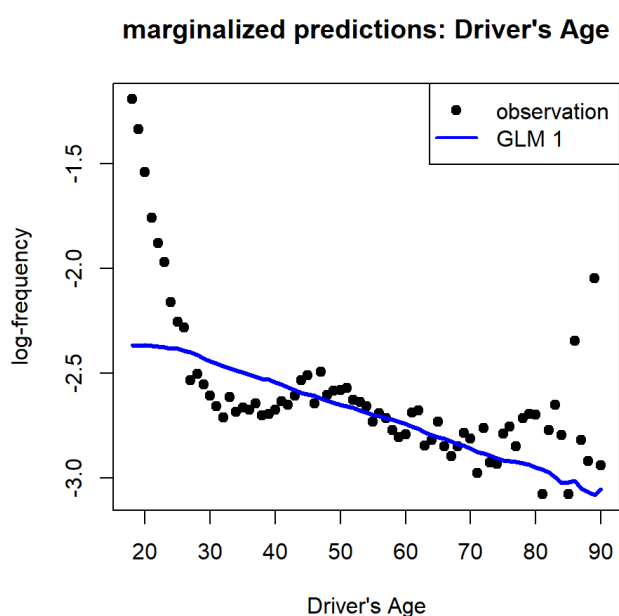
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.7811529	0.0298084	-93.30	<2e-16 ***
DrivAge	-0.0097160	0.0004414	-22.01	<2e-16 ***
lDensity	0.1023950	0.0032929	31.10	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

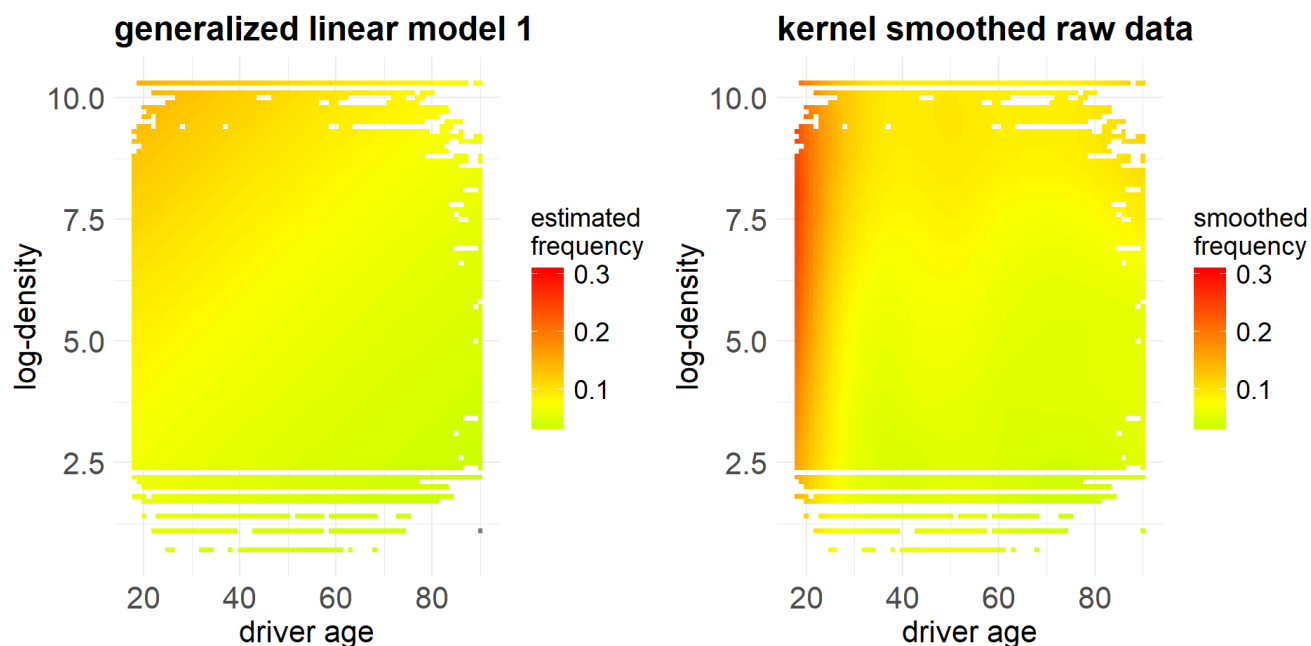
(Dispersion parameter for poisson family taken to be 1)

Null deviance: 9169.3 on 5988 degrees of freedom
Residual deviance: 7647.9 on 5986 degrees of freedom
AIC: 22487

Number of Fisher Scoring iterations: 5



- Obviously, the driver's age variable does not fit the observed data, as this GLM is not able to model a non-monotone structure.
- The log-Density modeling looks reasonable.



- The heatmap confirms that this GLM is not able to reflect the observed data.
- Non-monotonicity in `DrivAge` is an issue.

- In case of non-monotone behavior, one can build categorical classes.
- We build the age classes as follows (which are then implemented by dummy coding – explained in the Wednesday lecture).

```
### Model GLM 2
dat$DrivAgeGLM <- as.factor(cut(dat$DrivAge,
                                c(18,20,25,30,40,50,70,101), labels = c("18-20", "21-25", "26-30",
                                "31-40", "41-50", "51-70", "71+"), include.lowest = TRUE))

#
# Poisson GLM with offset and (canonical) log-link
d.glm2 <- glm(ClaimNb ~ DrivAgeGLM + lDensity,
              data=dat, offset=log(Exposure), family=poisson())

#
dat$GLM2 <- fitted(d.glm2)
```

- This estimates an individual regression parameter β_j for each age class.

Call:

```
glm(formula = ClaimNb ~ DrivAgeGLM + lDensity, family = poisson(),
     data = dat1, offset = log(Exposure))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.047914	0.044931	-45.58	<2e-16 ***
DrivAgeGLM21-25	-0.627963	0.046706	-13.45	<2e-16 ***
DrivAgeGLM26-30	-1.109290	0.045196	-24.54	<2e-16 ***
DrivAgeGLM31-40	-1.261688	0.042354	-29.79	<2e-16 ***
DrivAgeGLM41-50	-1.165688	0.042086	-27.70	<2e-16 ***
DrivAgeGLM51-70	-1.298062	0.041879	-31.00	<2e-16 ***
DrivAgeGLM71+	-1.380917	0.047348	-29.16	<2e-16 ***
lDensity	0.104394	0.003301	31.62	<2e-16 ***

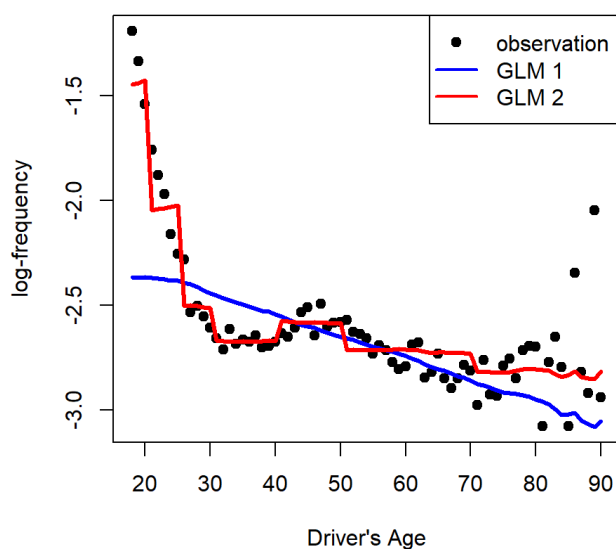
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

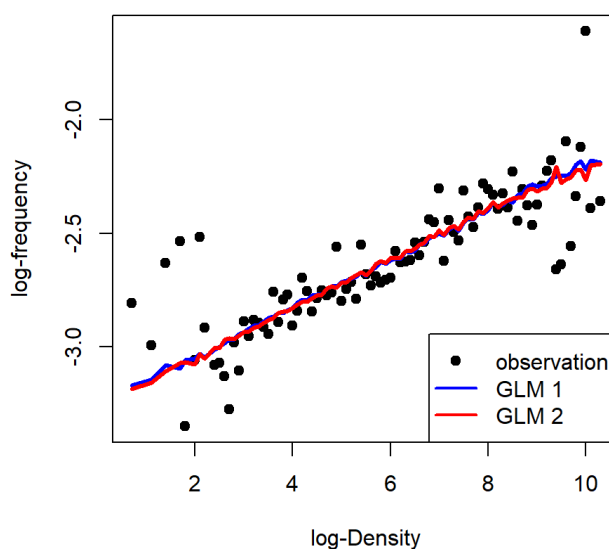
Null deviance: 9169.3 on 5988 degrees of freedom
Residual deviance: 6858.0 on 5981 degrees of freedom
AIC: 21707

Number of Fisher Scoring iterations: 5

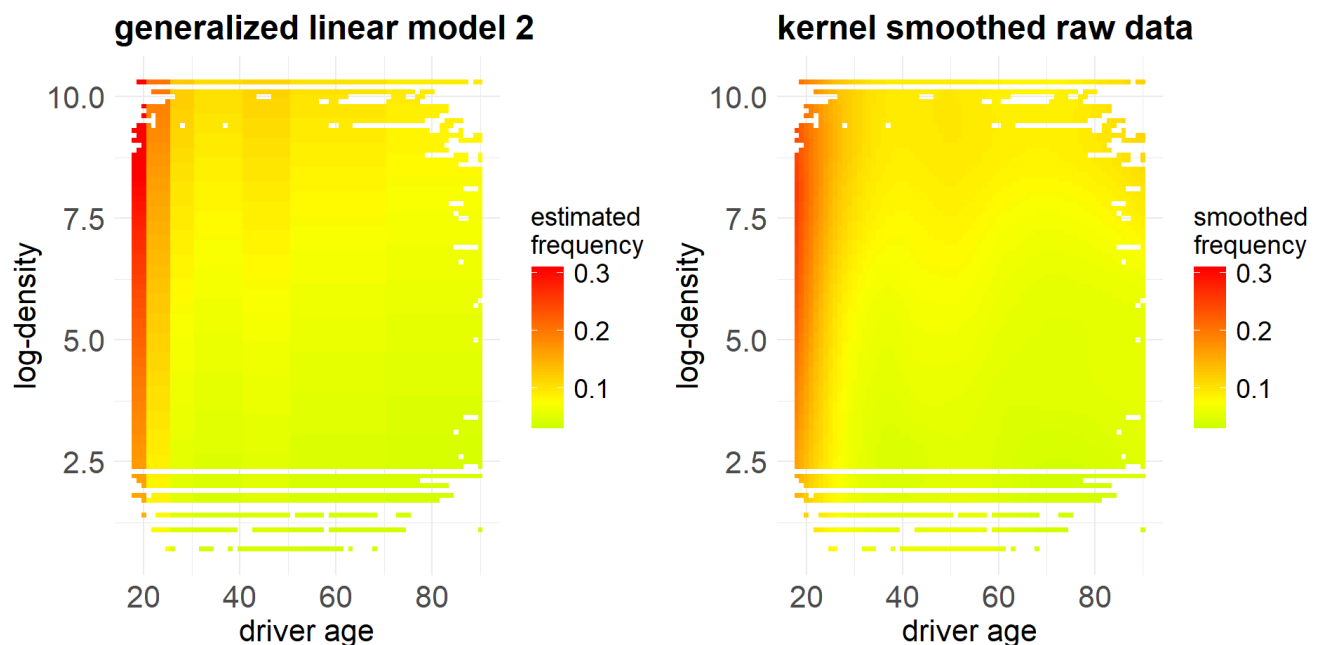
marginalized predictions: Driver's Age



marginalized predictions: log-Density



- This second GLM with categorical age classes fits the data better.
- Because of the log-link choice, it has a *multiplicative structure*, see next graph.

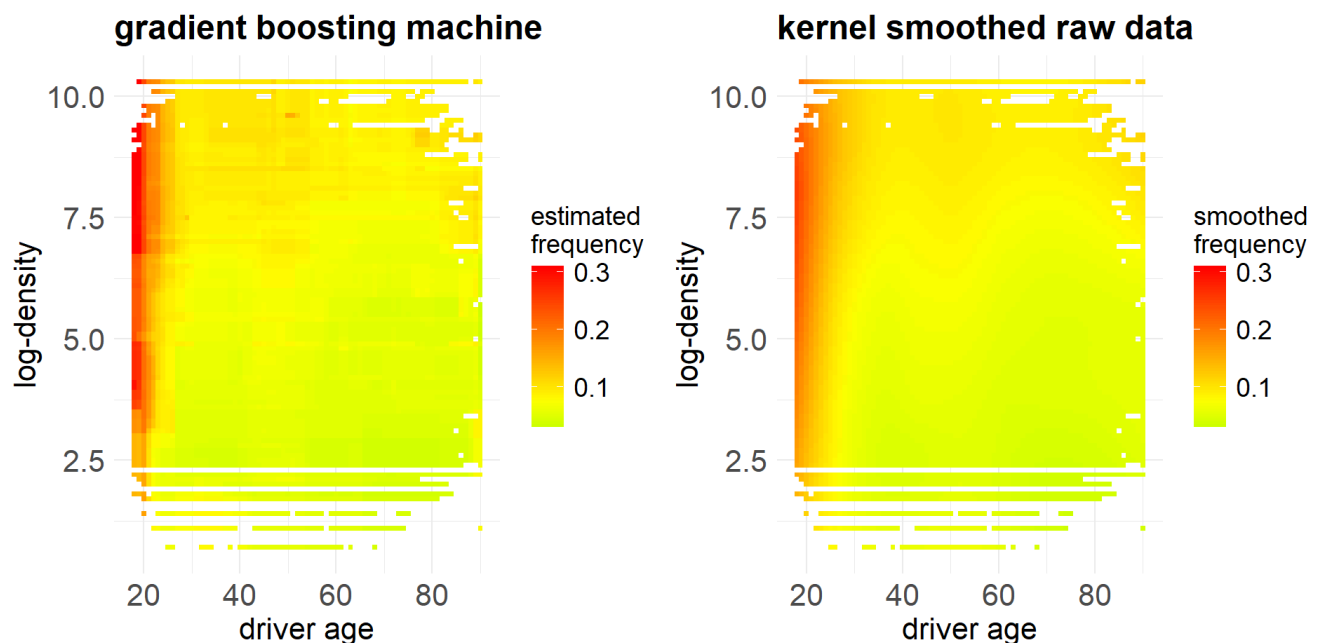


- This second GLM reflects the non-monotonicity in `DriveAge`.
- However, there is more (diagonal) structure which is a motivation to consider more complex regression models.

4 Gradient boosting machine

- As a first machine learning regression model, we present a *gradient boosting machine* (GBM).
- GBMs belong to the most powerful techniques on tabular data, in particular, the powerful versions like XGBoost, LightGBM or CatBoost.
- A GBM *partitions* the covariate space into different (homogeneous) subsets and it estimates the frequency on each subset.

- We only present the resulting heatmap on the two selected covariates; for technical details we refer to our ‘AI Tools for Actuaries’ homepage.
- GBMs are not further studied in this summer school.



- This result was generated by the LightGBM version.
 - We interpret this as follows.
-
- The GBM seems to reflect the observed (smoothed) data very well.
 - The only criticism that one may raise is that the resulting regression function is not smooth and that it cannot easily be extrapolated.
 - A crucial question that one may have is whether this GBM only extracts the structural (systematic) part or whether it also reflects some of the noise in the observed data. The latter would not be good, because the noisy part is not structure that will repeat in the future.
 - The previous issue is known as *in-sample over-fitting*.
 - In-sample over-fitting is a recurrent question to all statistical and machine learning methods, and it will be discussed in detail below.

- We are now ready to take off this process of statistical modeling and machine learning: please enjoy!

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https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5162304

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